MACHINE LEARNING ASSIGNMENT

K-NEAREST NEIGHBOURS (KNN)

Problem 1:

Consider the following two-dimensional dataset.

Point	x_1	x_2
A	2	3
B	5	4
C	1	1
D	6	7

The query point is

$$Q = (4, 5).$$

Calculate the distance between Q and each of the four points using:

(a) Euclidean distance

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

(b) Manhattan distance

$$d(x, y) = |x_1 - y_1| + |x_2 - y_2|$$

(c) Minkowski distance with $p = 3$

$$d(x, y) = \left(|x_1 - y_1|^3 + |x_2 - y_2|^3\right)^{1/3}$$

(d) Identify the nearest point under each distance measure.

Problem 2:

The following dataset contains students' marks in Mathematics and Statistics.

Student	Mathematics	Statistics	Class
A	80	75	Pass
B	65	70	Pass
C	40	45	Fail
D	35	50	Fail
E	72	68	Pass
F	50	55	Fail

A new student has marks

$$Q = (60, 60).$$

Using Euclidean distance:

(a) Calculate the distance from Q to every training observation.

(b) Arrange all observations in increasing order of distance.

(c) Classify Q using $K = 1$.

(d) Classify Q using $K = 3$.

(e) Classify Q using $K = 5$.

(f) Compare the three classifications.

Problem 3:

Consider the following one-dimensional dataset.

Observation	x	Class
A	1	Red
B	2	Red
C	3	Red
D	6	Blue
E	7	Blue
F	8	Blue
G	9	Blue

For the query point

$$Q = 5,$$

use KNN classification.

(a) Calculate the absolute distance from Q to every observation.

(b) Classify Q for $K = 1$.

(c) Classify Q for $K = 3$.

(d) Classify Q for $K = 5$.

(e) Classify Q for $K = 7$.

(f) Explain how increasing K changes the decision.

Problem 4:

Consider the following observations.

Point	x	Class
A	1	Red
B	2	Red
C	4	Blue
D	5	Blue

For the query point

$$Q = 3,$$

calculate the distances and perform KNN classification for:

(a) $K = 1$

(b) $K = 2$

(c) $K = 3$

(d) Discuss what happens when K is even and a tie occurs.

(e) Suggest a practical way to handle such a tie.

Problem 5:

Suppose a dataset contains two features:

$$\text{Age}$$

and

$$\text{Income.}$$

The observations are:

Person	Age	Income	Class
A	20	20000	No
B	25	25000	No
C	30	50000	Yes
D	35	60000	Yes
E	40	80000	Yes

A new person has

$$Q = (28, 52000).$$

(a) Explain why income may dominate age when Euclidean distance is calculated directly.

(b) Perform Min-Max normalization using

$$x' = \frac{x - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}}.$$

(c) Calculate the normalized features.

(d) Calculate the normalized distance from Q to each observation.

(e) Classify Q using $K = 3$.

Problem 6:

Consider a dataset with two features, X_1 and X_2.

For the two features, the following summary statistics are given:

$$\mu_1 = 50, \qquad \sigma_1 = 10$$

and

$$\mu_2 = 100, \qquad \sigma_2 = 20.$$

The observations are:

Point	X_1	X_2
A	40	80
B	50	100
C	60	120
D	70	140

A query observation is

$$Q = (55, 110).$$

Using standardization,

$$z = \frac{x - \mu}{\sigma},$$

calculate:

(a) The standardized coordinates of every observation.

(b) The standardized coordinates of Q.

(c) The Euclidean distance from Q to every observation.

(d) Identify the nearest observation.

Problem 7:

Consider the following training observations.

Point	x_1	x_2	Class
A	1	2	Red
B	2	3	Red
C	3	4	Blue
D	5	5	Blue
E	6	6	Blue

The query point is

$$Q = (3, 3).$$

Use $K = 3$ and Euclidean distance.

In weighted KNN, use

$$w_i = \frac{1}{d_i}.$$

(a) Calculate the distance from Q to every observation.

(b) Select the three nearest observations.

(c) Calculate the weight of each selected observation.

(d) Calculate the total weight for each class.

(e) Classify Q using weighted KNN.

(f) Compare the result with ordinary majority-voting KNN.

Problem 8:

KNN can also be used for regression.

Consider the following house-size and house-price data.

House	Size (m^2)	Price (lakhs)
A	800	40
B	1000	50
C	1200	60
D	1500	75
E	1800	90
F	2000	100

For a new house of size

$$Q = 1300\ \mathrm{m}^2,$$

use one-dimensional Euclidean distance.

(a) Find the distances from Q to all houses.

(b) Predict the price using KNN regression with $K = 1$.

(c) Predict the price using $K = 3$.

(d) Predict the price using $K = 5$.

(e) Explain why the predicted value changes as K changes.

For ordinary KNN regression, use

$$\hat{y} = \frac{1}{K}\sum_{i=1}^{K} y_i.$$

Problem 9:

Consider the following observations.

Point	x	y
A	2	20
B	4	30
C	5	40
D	7	50
E	9	70

For the query point

$$Q = 6,$$

use $K = 3$ and weighted KNN regression.

Use the weight formula

$$w_i = \frac{1}{d_i}.$$

The weighted prediction is

$$\hat{y} = \frac{\sum_{i=1}^{K} w_i y_i}{\sum_{i=1}^{K} w_i}.$$

(a) Calculate all distances.

(b) Identify the three nearest observations.

(c) Calculate their weights.

(d) Calculate the weighted prediction.

(e) Calculate the ordinary KNN regression prediction.

(f) Compare the two predictions.

Problem 10:

Consider the following classification problem.

Point	X_1	X_2	X_3	Class
A	1	2	1	A
B	2	2	1	A
C	3	3	2	A
D	6	6	5	B
E	7	5	6	B
F	8	7	7	B

The query point is

$$Q = (4, 4, 3).$$

Using Euclidean distance,

$$d(x, q) = \sqrt{(x_1 - q_1)^2 + (x_2 - q_2)^2 + (x_3 - q_3)^2},$$

perform the following:

(a) Calculate the distance from Q to every observation.

(b) Rank all observations from nearest to farthest.

(c) Classify Q using $K = 1$.

(d) Classify Q using $K = 3$.

(e) Classify Q using $K = 5$.

(f) Explain whether increasing K makes the classifier more or less sensitive to individual observations.

(g) Explain the effect of high dimensionality on distance-based methods.

(h) What is meant by the "curse of dimensionality"?

Conceptual Review

Answer the following questions briefly.

1. What is K-Nearest Neighbours?
2. Why is KNN called a lazy learning algorithm?
3. What is meant by a feature space?
4. Why is distance important in KNN?
5. Write the formula for Euclidean distance.
6. Write the formula for Manhattan distance.
7. What is the role of K in KNN?
8. What happens when K is very small?
9. What happens when K is very large?
10. Why is feature scaling important in KNN?
11. What is weighted KNN?
12. What is KNN regression?
13. What is the curse of dimensionality?
14. Mention two advantages of KNN.
15. Mention two disadvantages of KNN.